A Verification Method for Game Image AssetsUsing C2PA Manifest Validation and Perceptual Hash-Based Similarity Analysis

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1. Motivation and Problem

1.1 Background and Research Question

Motivation. Game assets are repeatedly re-saved or transformed during collaboration, web upload, and messenger sharing. Provenance may change even when visual content remains recognizable.

Research Question. Can C2PA verify an original first, and can perceptual hashing recover its source after provenance loss?

Experimental Setup. The dataset contains 50 originals, 250 controls, and 850 transformed images across 17 realistic conditions.

Transformation Conditions. Compression, resizing, cropping, color adjustment, format conversion, metadata removal, watermarking, screenshots, and rotation were evaluated.
Study Scope. The method verifies whether a query is original or derived from a known game image asset.

Table 1. Experimental dataset and evaluation scale.

Dataset	Images	Purpose
Original assets	50	C2PA-signed source set
Control images	250	Five file-processing conditions
Transformed images	850	Seventeen transformations
C2PA targets	*1,150*	Manifest validation

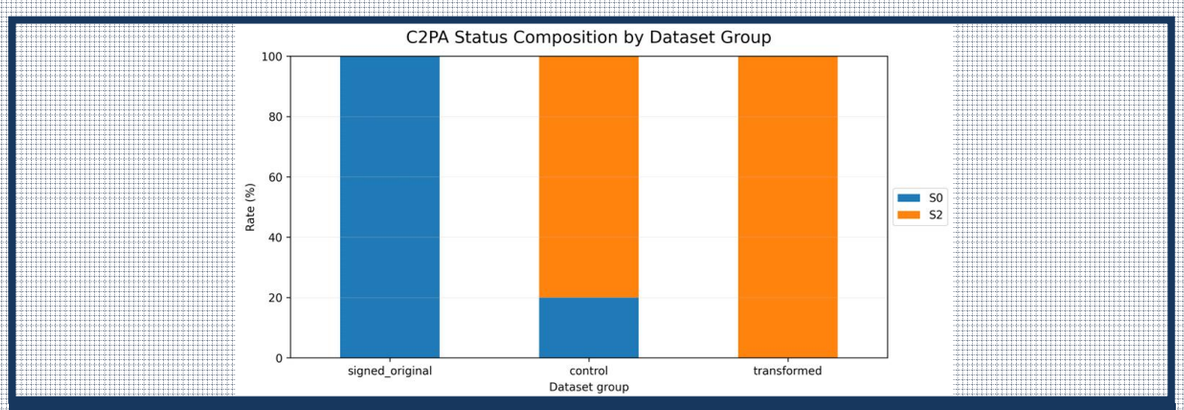


Fig. 1. C2PA status distribution by dataset group.

2. Second Section

2.1 Manifest Preservation

Preservation. All 50 signed originals and 50 byte copies retained valid claims; all 850 transformed images lost the claim.

Interpretation. C2PA is strong primary evidence but is fragile in ordinary processing pipelines.

2.2 Practical Implications and Limitations

Operational Implication. C2PA can verify intact signed files, but ordinary image tools may remove the Manifest during re-encoding or metadata processing.

Fallback Evidence. Perceptual hashing can recover the correct source candidate when compression, resizing, color adjustment, conversion, metadata removal, or watermarking preserves global structure.

Limitations. Screenshots, substantial cropping, and rotations of 15° or more remain unreliable and should not be assigned automatically.

Practical Use. A verifier should combine primary provenance evidence, supplementary similarity evidence, and manual review for ambiguous cases.

3. Proposed Verification Method

3.1 Two-Stage Verification Procedure

Stage 1 - C2PA Validation. A valid claim, signature, and data hash provide primary evidence.

Stage 2 - Hash Retrieval. If no claim remains, four hashes rank 50 originals by Hamming distance.

Similarity Metric. A smaller 64-bit Hamming distance indicates a more perceptually similar candidate.

Decision. A valid claim indicates verified originality; a confident hash match indicates a candidate derivative; uncertain cases require review.

4. Perceptual-Hash Results

4.1 Identification Accuracy

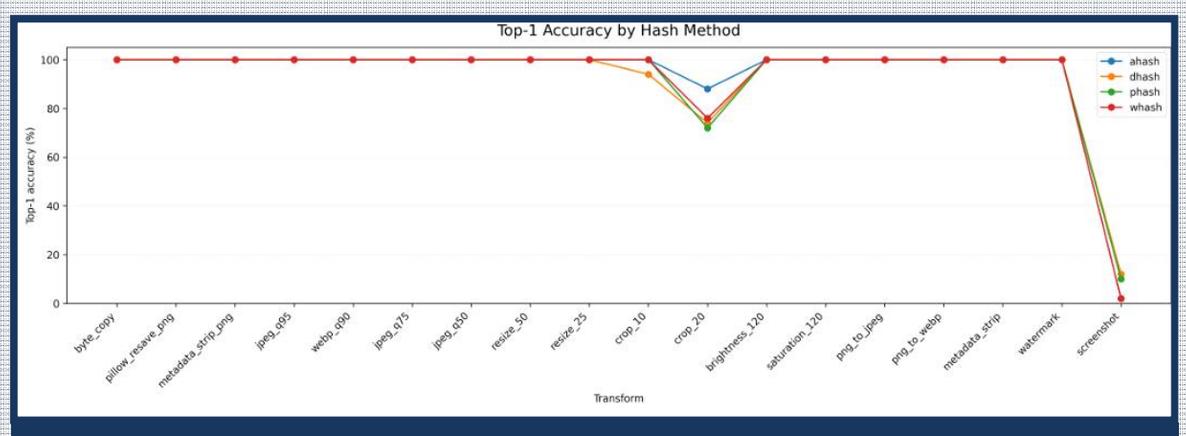


Fig. 2. Top-1 accuracy by hash method and transformation condition.

Robust Conditions. All hashes reached 100% under JPEG Q95, 50% resizing, metadata removal, and watermarking.

Difficult Conditions. Crop 20% achieved 74-88%; screenshots achieved 0-8%; rotations of 15° or more achieved at most 12%.

Rotation Result. pHash performed best at 5° rotation with 82% Top-1 accuracy.

Interpretation. Whole-image hashes remain effective while global visual structure is preserved.
Review Rule. Structurally difficult cases should remain unknown until manually reviewed.

5. Conclusion

5.1 Practical Verification Policy

Primary Evidence. Verify valid C2PA claims before similarity analysis.

Supplementary Evidence. Use perceptual hashing only after claim loss.

Escalation. Defer screenshots, large rotations, and ambiguous crops for review.

PRIMARY EVIDENCE + SUPPLEMENTARY EVIDENCE
A layered verifier combines both.

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